

Lezione 2 – Studi di funzione

Matematica del continuo

Modulo 11 - Studio di funzioni

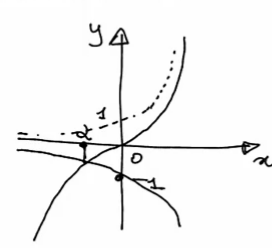
Unità didattica 4 – Funzioni iperboliche

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$$y = \frac{\operatorname{Sh} x}{\operatorname{Sh} x + e^x}$$

$\operatorname{Sh} x + e^x = 0 \quad \operatorname{Sh} x = -e^x$



CE: $(-\infty, 2) \cup (2, +\infty)$

$$\frac{e^x - e^{-x}}{2} + e^x = 0$$

$$e^x - e^{-x} + 2e^x = 0$$

$$3e^x - e^{-x} = 0$$

$$3e^{2x} = 1$$

$$e^{2x} = 1/3$$

$$2x = \log \frac{1}{3} = -\log 3$$

$$x = -\frac{1}{2} \log 3$$

$$\begin{array}{l}
 y(0) = 0 \\
 y > 0 \text{ per } x < \alpha \text{ e } x > 0 \\
 y < 0 \text{ per } \alpha < x < 0
 \end{array}
 \quad \left| \quad \alpha = -\frac{1}{2} \log 3
 \right.$$

$$CE \left(-\infty, -\frac{1}{2} \log 3 \right) \left(-\frac{1}{2} \log 3, +\infty \right)$$

$$\lim_{x \rightarrow -\infty} y = 1 \quad \left\{ \begin{array}{l} -\infty \\ -\infty + \underbrace{e^{-\infty}}_{0^+} \end{array} \right.$$

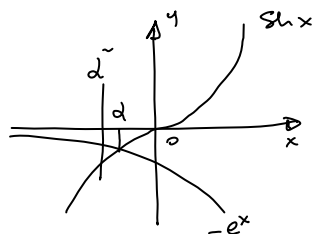
$$\text{la retta } y = 1 \text{ e}^-$$

$$\text{As. Orizz per } x \rightarrow -\infty$$

$$\lim_{x \rightarrow +\infty} y = \frac{1}{3} \quad \frac{\text{Sh } x}{\text{Sh } x + e^x} \quad \left(\frac{e^{x/2}}{3e^{x/2}} \right)$$

$$\text{la retta } y = 1/3 \text{ e}^-$$

$$\text{As. Orizz per } x \rightarrow +\infty$$



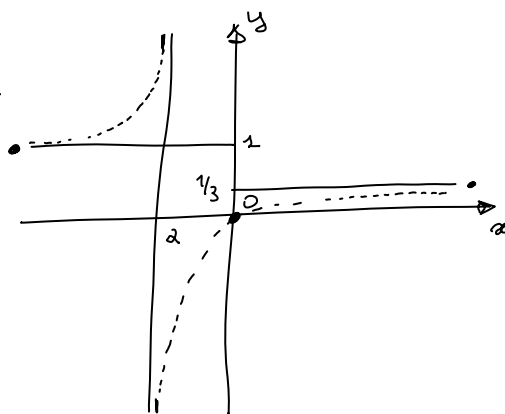
$$\alpha = -\frac{1}{2} \log 3$$

$$\lim_{x \rightarrow \alpha^-} y = +\infty \quad \frac{\text{neg}}{0^+}$$

$$\lim_{x \rightarrow \alpha^+} y = -\infty$$

$$x = \alpha \text{ e un As Vert}$$

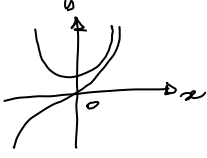
$$\text{Si } e^x \text{ che } e^{-x} \text{ che } e^{x/2}$$



$$y = \frac{\operatorname{sh} x}{\operatorname{sh} x + e^x}$$

$$y' = \frac{\operatorname{ch} x (\operatorname{sh} x + e^x) - \operatorname{sh} x (\operatorname{ch} x + e^x)}{(\operatorname{sh} x + e^x)^2} = \frac{e^x (\operatorname{ch} x - \operatorname{sh} x)}{(\operatorname{sh} x + e^x)^2}$$

$y' > 0$ funzione crescente



$$y'' = \frac{1}{(\operatorname{sh} x + e^x)^3} \left[(e^x (\operatorname{ch} x - \operatorname{sh} x) + e^x (\operatorname{sh} x - \operatorname{ch} x)) (\operatorname{sh} x + e^x)^2 - e^x (\operatorname{ch} x - \operatorname{sh} x) 2 (\operatorname{sh} x + e^x) (\operatorname{ch} x + e^x) \right]$$

$$= \frac{-e^x}{(\operatorname{sh} x + e^x)^3} (\operatorname{ch} x - \operatorname{sh} x) (\operatorname{ch} x + e^x)$$

$\underbrace{(\operatorname{ch} x - \operatorname{sh} x)}_{>0}$

FINE

